

Experiment 2

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Subject Name: Technical Training

Subject Code: 25CAP-652

Title: Designing and implementing a sample database system using DDL, DML, and DCL commands.

1. Aim:

- a. To design and implement a sample database system using DDL, DML, and DCL commands, including database creation, data manipulation, schema modification, and role-based access control to ensure data integrity and secure, read-only access for authorized users.

2. Software Requirements:

- a. PostgreSQL (Database Server)
- b. pgAdmin
- c. OS: Windows

3. Objective:

- a. To gain practical experience in implementing Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL) operations in a real database environment. This will also include implementing role-based privileges to secure data.

4. Procedure of the Practical:

- a. Start the system.
- b. Open pgAdmin.
- c. Create and select the database in which you want to perform the experiment.
- d. Establish connection to the database using Alt+Shift+Q.
- e. Run the following queries given in Experiment Steps (5).

5. Experiment Steps:

a. Create table queries:

- i. Create table department (dept_id int primary key, dept_name varchar (20) not null unique);
- ii. Create table project (proj_id int primary key, proj_name varchar (50) not null, dept_id int not null, foreign key (dept_id) references department(dept_id));
- iii. Create table employee (emp_id varchar (10) primary key, emp_name varchar (30) not null, dept_id int not null, proj_id int not null, foreign key (dept_id) references department (dept_id) on delete cascade on update cascade, foreign key (proj_id) references project (proj_id) on delete cascade on update cascade);

b. Insert queries:

- i. Insert into department values (1, 'Sales'), (2, 'Marketing'), (3, 'Legal'), (4, 'IT'), (5, 'Accounts'), (6, 'HR');

	dept_id [PK] integer	dept_name character varying (20)
1	1	Sales
2	2	Marketing
3	3	Legal
4	4	IT
5	5	Accounts
6	6	HR

- ii. Insert into project values (1000, 'lead gen campaign', 1);

	proj_id [PK] integer	proj_name character varying (50)	dept_id integer
1	1000	Lead Gen Campaign	1
2	1001	CRM CleanUP	1
3	2000	Content Campaign	2
4	2001	Influencer Marketing	2
5	3000	Contract Audits	3
6	3001	Litigation Tools	3
7	4000	Cloud Migration	4
8	4001	Cybersecurity Setup	4
9	5000	Financial Audits	5
10	5001	Forecasting Models	5
11	6000	Wellness Programs	6
12	6001	Onboarding Surveys	6

iii. Insert into employee values ('E100', 'Vikram Singh', 1, 1000);

	emp_id [PK] character varying (10)	emp_name character varying (30)	dept_id integer	proj_id integer
1	E100	Vikram Singh	1	1000
2	E101	Ritu Das	5	5000
3	E102	Sonia Patel	6	6001
4	E103	Kavita Sharma	6	6001
5	E104	Arjun Gupta	1	1000

c. DML UPDATE and DELETE

i. Update employee set proj_id = 6000 where proj_id = 6001;

18	E102	Sonia Patel	6	6000
19	E103	Kavita Sharma	6	6000
20	E107	Neha Gupta	6	6000

ii. Delete from project where dept_id = 3;

	proj_id [PK] integer	proj_name character varying (50)	dept_id integer
1	1000	Lead Gen Campaign	1
2	1001	CRM CleanUP	1
3	2000	Content Campaign	2
4	2001	Influencer Marketing	2
5	4000	Cloud Migration	4
6	4001	Cybersecurity Setup	4
7	5000	Financial Audits	5
8	5001	Forecasting Models	5
9	6000	Wellness Programs	6
10	6001	Onboarding Surveys	6

d. Access Control and Security

i. Creating a user: Create user reporter with password 'reporter@123';

```
CREATE ROLE
```

```
Query returned successfully in 37 msec.
```

ii. Granting select privileges: Grant select on employee, department, project to reporter;

```
GRANT
```

```
Query returned successfully in 31 msec.
```

- iii. Explicitly revoking create privilege on the DB: Revoke create on schema public from reporter;

```
REVOKE
```

```
Query returned successfully in 29 msec.
```

e. Schema modification

- i. Alter table: Alter table employee add column base_sal int;

	emp_id [PK] character varying (10)	emp_name character varying (30)	dept_id integer	proj_id integer	base_sal integer
1	E100	Vikram Singh	1	1000	[null]
2	E101	Ritu Das	5	5000	[null]
3	E104	Arjun Gupta	1	1000	[null]
4	E105	Sonia Gupta	1	1000	[null]
5	E108	Vikram Patel	1	1000	[null]

- ii. Alter table to remove foreign key constraint further to drop project table:

Alter table employee drop constraint employee_proj_id_fkey;

> department

✓ employee

> Columns

✓ Constraints (2)

employee_dept_id_fkey

employee_pkey

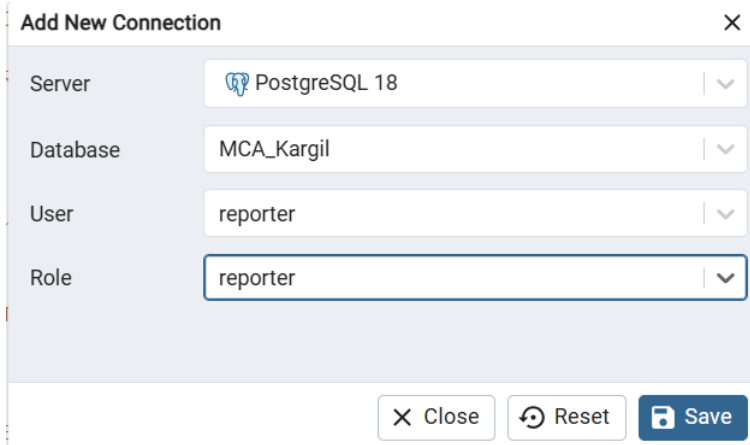
- iii. Drop table: Drop table project;

```
102 select * from project;
```

Data Output Messages Notifications

```
ERROR: relation "project" does not exist
LINE 102: select * from project;
```

f. Creating new connection to test created role:



Add New Connection [X]

Server: PostgreSQL 18 [v]

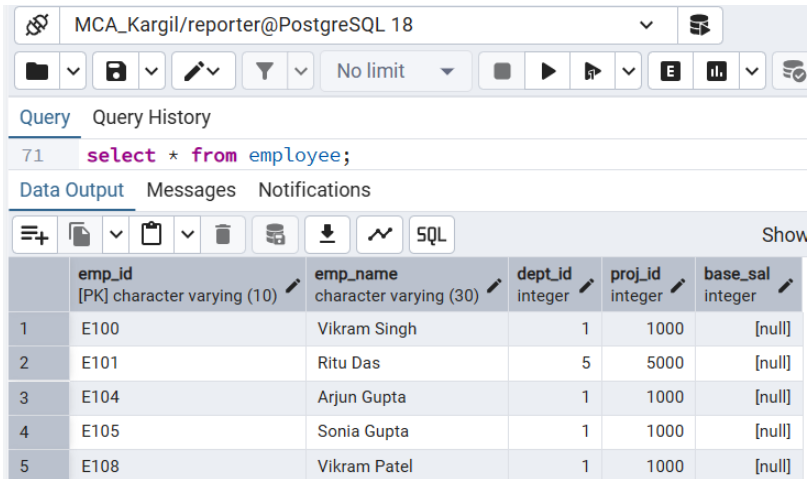
Database: MCA_Kargil [v]

User: reporter [v]

Role: reporter [v]

[X Close] [Reset] [Save]

g. Checking if role “reporter” has select privilege on employee relation:



MCA_Kargil/reporter@PostgreSQL 18

Query: `select * from employee;`

Data Output: Messages Notifications

	emp_id [PK] character varying (10)	emp_name character varying (30)	dept_id integer	proj_id integer	base_sal integer
1	E100	Vikram Singh	1	1000	[null]
2	E101	Ritu Das	5	5000	[null]
3	E104	Arjun Gupta	1	1000	[null]
4	E105	Sonia Gupta	1	1000	[null]
5	E108	Vikram Patel	1	1000	[null]

h. Check if role “reporter” has create privileges on ‘Experiment_1’ schema:



MCA_Kargil/reporter@PostgreSQL 18

Query: `create table test_table(id int);`

Data Output: Messages Notifications

ERROR: permission denied for schema Experiment_1
LINE 24: create table test_table(id int);
^

6. Input/Output Details:

- a. The input for the experiment is the SQL queries mentioned in Experiment Steps (5).
- b. The output for each query is attached after the query itself.

7. Learning Outcome

- a. Learning DDL, DML, and DCL commands in SQL.
- b. How to implement integrity and domain constraints in SQL.
- c. DBMS privilege management.